FIELD PROJECT REPORT

(Semester III – B.Sc. Data Science)



University of Mumbai

PROJECT PARTICULARS

ANALYSIS OF
SEASONAL DISEASE PATTERNS
USING GOVERNMENT HEALTH DATA

A study of NCDC / IDSP outbreak surveillance records in Maharashtra

DATA • SEASONALITY • PUBLIC HEALTH



SUBMITTED BY

Khan Umar Abdulaziz

ROLL NUMBER	41
CLASS	S.Y. B.Sc. Data Science
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COLLEGE	R.P. Institute of Hospitality and Management
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PROJECT GUIDE	Prof. Angelin

UNIVERSITY OF MUMBAI | FIELD PROJECT REPORT

Academic submission • Maharashtra, India

R. P. Institute of Hospitality & Management

University of Mumbai

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Certificate

Field Project Report • Academic Year 2026–2027

This is to certify that Mr. Khan Umar Abdulaziz, Roll No. 41, a student of S.Y. B.Sc. (Data Science), Semester III, at R.P. Institute of Hospitality and Management, Vasai, affiliated to the University of Mumbai, has submitted the field project entitled:

“Analysis of Seasonal Disease Patterns Using Government Health Data”

The project examines government-notified outbreak surveillance records for Maharashtra and presents a descriptive analysis of seasonal distribution, concentration, district variation, data quality, and methodological limitations. It is submitted in partial fulfilment of the requirements for the Semester III field project in the B.Sc. (Data Science) programme during the academic year 2026–2027.

This certificate is presented for review and institutional approval. The Project Guide and Head of Department are requested to sign in the designated spaces, with the official college stamp placed in the centre.

_____ **Prof. Angelin** **Project Guide**	COLLEGE STAMP	_____ Prof. Khushali Sharma Head of Department

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Declaration

I, Khan Umar Abdulaziz, a student of S.Y. B.Sc. Data Science, Semester III, Roll No. 41, at R.P. Institute of Hospitality and Management, hereby declare that this field project report titled “Analysis of Seasonal Disease Patterns Using Government Health Data” has been prepared for academic submission during the academic year 2026–2027.

The quantitative analysis presented in this report is based on the documented NCDC/IDSP outbreak-surveillance archive and the analytical outputs stored in the project repository. The report distinguishes source data, derived calculations, interpretation, and limitations. Where repository summaries conflict with the family-level dataset totals, the discrepancy is explicitly disclosed rather than concealed.

The fieldwork archive contains questionnaires, an interview guide, an observation checklist, an ethics protocol, and a planned analysis framework. The evidence register currently contains only its header and no completed entries; therefore, I have not represented planned interviews, questionnaire responses, site observations, or photographs as completed empirical fieldwork.

I have acknowledged the data sources and project artefacts used. Any final statement of originality, completion, and institutional approval should be confirmed by the student and the authorised college signatories before submission.

Student Signature: _____

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Date: October 2026

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Acknowledgement

I would like to express my sincere gratitude to Prof. Angelin, my project guide, for her guidance, encouragement, and support throughout the planning and development of this field project. Her guidance helped me approach the subject with care and focus on evidence-based analysis.

I am thankful to Prof. Khushali Sharma, the Head of Department, for her encouragement and academic support. I also acknowledge R.P. Institute of Hospitality and Management, Vasai, and the University of Mumbai for providing the academic setting in which this project was undertaken.

I am grateful for the public health surveillance records and documentation made available through the National Centre for Disease Control and the Integrated Disease Surveillance Programme. These sources made it possible to study reported outbreak patterns across Maharashtra and to build a documented data-processing and analysis workflow.

I would also like to thank my classmates, friends, and family for their encouragement during the completion of this work. Finally, I acknowledge the importance of transparent data handling: this report separates completed secondary-data analysis from field instruments for which completed primary evidence has not been documented in the repository.

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Abstract

This field project investigates seasonal patterns in government-reported disease outbreaks across Maharashtra, India, using weekly outbreak surveillance reports attributed to the National Centre for Disease Control (NCDC) and the Integrated Disease Surveillance Programme (IDSP). The main completed analytical baseline covers the four full calendar years 2022–2025 and includes 539 reported outbreak records across five selected disease families: dengue, acute diarrheal disease (ADD), malaria, food poisoning, and chikungunya. The family-level analytical tables sum to 20,955 reported cases and 266 reported deaths. A separate 2026 comparison covers epidemiological Weeks 1–32 only and is treated as a partial, out-of-sample period rather than as a complete year.

The repository documents a staged workflow covering extraction from weekly PDF reports, row-level verification, disease-label normalization, monthly aggregation, seasonal-index calculation, concentration metrics, peak-window analysis, statistical testing, outlier sensitivity analysis, and district-level comparisons. The findings show a late-year concentration in dengue reporting, a June–August elevation for ADD, and a recurring May–July window for malaria. Food-poisoning case totals are highly sensitive to a small number of large reported events, while chikungunya records are comparatively sparse and temporally diffuse. Eastern Vidarbha accounts for a substantial share of malaria outbreak notifications.

The report also identifies discrepancies between some narrative summary totals and the disease-family rows in the repository, including baseline and partial-2026 case totals. These are recorded as data-reconciliation issues. The fieldwork archive contains questionnaires, interview guidance, observation checklists, and ethics procedures, but its evidence register has no completed entries; therefore, this report does not claim to have empirical interview or survey results. The findings are descriptive of reported outbreak surveillance and do not estimate population incidence, individual risk, or causal effects.

Keywords: seasonal disease patterns; NCDC; IDSP; Maharashtra; outbreak surveillance; seasonal index; data quality; disease reporting.

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539	20,955	266	5
Baseline outbreak records 2022–2025	Reported cases family-level sum	Reported deaths family-level sum	Primary disease families comparative analysis

The 2026 Weeks 1–32 cohort is treated separately from completed years; summary-total discrepancies are documented in Chapter 4.

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List of Tables

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Abbreviations

Abbreviation	Meaning
ADD	Acute Diarrheal Disease (repository-normalized family label)
CV	Coefficient of Variation
CR3	Share of records/cases in the three highest months
IDSP	Integrated Disease Surveillance Programme
NCDC	National Centre for Disease Control
YTD	Year to Date
KW	Kruskal–Wallis test
FDR	False Discovery Rate
PII	Personally Identifiable Information

CHAPTER 1

Introduction and Background

1.1 Background and Context

Seasonal disease surveillance is important because the timing of reported outbreaks affects when public-health teams prepare staff, diagnostic supplies, vector-control activity, risk communication, and water- and food-safety responses. Maharashtra contains urban, peri-urban, rural, coastal, and inland settings; a state-wide summary can therefore mask local patterns. The purpose of this project is not to declare a single statewide “disease season,” but to examine how government-notified outbreak records are distributed over months and years and how stable those distributions appear.

The study uses weekly outbreak reports attributed to NCDC and IDSP. These reports record notified outbreak events or clusters, with disease labels, reporting period, geography, reported cases, and deaths where supplied. An outbreak record is not equivalent to an individual patient record, and a sum of reported outbreak cases is not the same as a state population incidence rate. This distinction governs the interpretation throughout the report.

The repository documents an end-to-end workflow: source-PDF archiving, extraction, validation, normalization of inconsistent disease labels, monthly and weekly aggregation, seasonal indices, concentration measures, statistical tests, outlier sensitivity checks, and an evidence-synthesis layer. It also provides a website intended to communicate the results and a separate fieldwork archive that describes possible questionnaires, interviews, and observation methods.

The academic focus is a data-science investigation of a public-health question: what recurring temporal patterns are visible in official outbreak notifications, what remains unstable, and what limitations should a reader keep in mind?

CHAPTER 1

Problem Statement, Objectives and Research Questions

1.2 Problem Statement

Weekly outbreak surveillance tables are published as reports and may contain changing labels, reporting gaps, unusual high-volume events, and differences in district reporting. If these records are simply summed without validation, a few extreme outbreaks or a partial reporting year can distort apparent seasonality. This project addresses the need for a transparent, reproducible description of outbreak-reporting patterns and an explicit record of what the data can and cannot support.

Aim

To analyze seasonal patterns in NCDC/IDSP government-reported outbreak records for selected disease families in Maharashtra, using validated data processing and descriptive/statistical methods.

Objectives

- To organize and validate weekly outbreak records for the completed 2022–2025 baseline and keep partial 2026 records separate.
- To standardize source disease labels into five selected analytical families: dengue, ADD, malaria, food poisoning and chikungunya.
- To describe monthly and weekly timing using event counts, seasonal indices and three-month peak windows.
- To compare seasonal concentration, inter-annual stability, district concentration and event-versus-case burden.
- To test sensitivity to influential records and document inconsistencies in narrative summaries.
- To prepare transparent fieldwork instruments and an ethical plan for possible contextual inquiry without claiming unrecorded fieldwork as completed.

Research Questions

- Which months contain the largest share of reported outbreak events for each selected disease family?
- Which patterns recur across 2022–2025, and which vary between years or districts?
- How strongly do case totals respond to a small number of high-volume outbreak records?
- What does the partial 2026 cohort show, and why must it remain separate from full-year comparisons?

CHAPTER 2

Review of Literature and Data-Source Context

2.1 Seasonal Disease Surveillance

Seasonal patterns in public-health data are shaped by multiple interacting processes: exposure conditions, pathogen and vector ecology, human behaviour, healthcare access, reporting practices, outbreak definitions, and the timing of surveillance publication. A monthly peak in notified outbreaks can be epidemiologically meaningful, but the pattern alone does not prove that temperature, rainfall, sanitation, or any other single factor caused the peak. The correct unit of interpretation in this project is the reported outbreak record.

The project repository includes a Maharashtra-specific study titled “The Demographic and Seasonal Variation of Dengue Virus Infection in Gondia, A Tribal District in Maharashtra, India” (listed in the repository source audit as an IJCPR paper, 2026). This source provides contextual evidence that local dengue activity can display a seasonal profile, but its clinical-confirmed case series is not identical to the statewide IDSP outbreak-notification series used in this report. Direct numerical comparisons should therefore be cautious.

The repository also identifies a 2016 literature item on seasonal trends in IDSP outbreak data. Its role is background support for the use of routine surveillance records and disease-family comparisons; this report does not import national aggregates from that source into the Maharashtra baseline.

2.2 Why Official Weekly Reports Were Selected

The source audit in the repository records that weekly NCDC/IDSP outbreak reports contain date/week, state/district, disease label, case count, and death count for notified clusters. Weekly resolution allows monthly aggregation and approximate within-year timing, while district labels allow a basic spatial concentration assessment. Annual tables can be useful for national or state summaries but are insufficient on their own for a month-by-month analysis.

CHAPTER 2

Source Audit, Conceptual Framework and Research Gap

2.3 Source and Dataset Audit

Repository asset	Role in this report	Interpretive note
NCDC weekly outbreak PDFs, 2022–2025	Primary source archive for baseline extraction	Published outbreak tables; not person-level surveillance microdata.
NCDC weekly PDFs, 2026 Weeks 1–32	Partial out-of-sample comparison	The post-monsoon/autumn window is incomplete.
NCDC_Maharashtra_MASTER_RAW.csv (809 records)	Extracted master record set	Includes more than the five primary families.
NCDC_Maharashtra_CLEAN_ANALYTICAL.csv	Clean analytical table	Repository reports row-level audit trail; use selected family cohort for the main totals.
Phase 5B–5D reports and CSV outputs	Seasonality, statistics, interpretation and limitations	Summary files contain a few unresolved total discrepancies; see Chapter 4.
Field-work instruments and evidence register	Primary-fieldwork readiness and ethics	Register is empty; no empirical interview/survey claims are made.

2.4 Conceptual Framework

The analytical logic is: official weekly reports → extracted records → field and disease labels checked → normalized disease families → completed-year baseline separated from partial 2026 → monthly/weekly summaries → concentration and statistical tests → sensitivity and district analysis → evidence-graded conclusions. A parallel fieldwork path is planned to collect context (perceptions, notification pathways, and environmental observations) only if genuine evidence is subsequently obtained and documented.

Research Gap

The practical gap addressed here is not the absence of all public-health information, but the need to convert scattered weekly reports into a transparent multi-year view that distinguishes repeated temporal structure from artifacts caused by incomplete years, inconsistent labels, district concentration, and exceptional mass-exposure events.

CHAPTER 3

Research Methodology

3.1 Research Design and Study Area

The completed component is a quantitative, descriptive and analytical secondary-data study of government-reported outbreak notifications in Maharashtra, India. The main baseline is the four completed calendar years 2022–2025. Partial 2026 records through epidemiological Week 32 are treated as an out-of-sample surveillance window and excluded from full-year seasonal-index computation.

Dimension	Specification
Geographic unit	Maharashtra, state-level summaries with district-level concentration checks
Primary period	2022–2025 completed calendar years
Secondary period	2026, Weeks 1–32 only; partial and separate
Observation unit	One government-notified outbreak record, not one individual patient
Selected families	Dengue; Acute Diarrheal Disease (ADD); Malaria; Food Poisoning; Chikungunya
Main outputs	Monthly and weekly distributions, seasonal indices, concentration measures, year stability, district patterns, sensitivity tests

3.2 Selection of Disease Families

Phase 5A documented the normalization of 49 raw disease labels into 30 analytical families, from which five primary families were selected for this project based on repository-defined data availability and public-health relevance. In 2026, label shifts remain visible: ADD records use “Diarrhoeal,” while food-poisoning records use “Suspected Food Poisoning.” A normalized family label is useful for aggregation, but it does not remove uncertainty in the underlying clinical classification.

No population denominator is available in the main analytical table. Therefore, the study does not calculate incidence per 100,000 people, community prevalence, attack rates, or person-level risk.

CHAPTER 3

Data Acquisition, Cleaning and Validation

3.3 Pipeline Steps

1. Source control: organize weekly NCDC reports by year and epidemiological week; preserve original reports as provenance.
2. Extraction: collect outbreak records and capture disease label, year/week, date, district, cases and deaths where shown.
3. Validation: compare extracted values with the source PDF; preserve row-level identifiers and audit status.
4. Normalization: map raw disease terms into standardized disease families and flag “suspected”/spelling variants rather than silently treating them as lab-confirmed diagnoses.
5. Time derivation: extract month from the event date/week fields using the repository’s analytical mapping; produce monthly and weekly counts.
6. Baseline isolation: restrict the main seasonal profile to the 539 selected records from full years 2022–2025; isolate the 45 selected 2026 records through Week 32.
7. Interpretation: cross-check monthly counts against concentration, stability, outlier and district-level analyses before assigning evidence tiers.

3.4 Validation Cautions Recorded in the Repository

The repository notes missing 2023 weekly report files for Weeks 15, 51 and 52; the analysis does not invent zero-outbreak records for missing weeks. A separate concern is a December 2025 ADD mortality cluster in which three entries show identical high fatality counts alongside very small case counts. The rows are preserved for provenance, but the counts are not interpreted as evidence of unusual pathogen lethality. Such exceptions motivate the sensitivity and reconciliation steps described later.

The raw/master extraction contains 809 records across the wider 2022–2026 archive; 539 records form the chosen five-family 2022–2025 baseline. These counts answer different questions and should not be conflated.

CHAPTER 3

Analytical Measures and Statistical Tests

3.5 Descriptive Measures

Measure	Meaning in this study
Monthly event count	Number of reported outbreak records assigned to each calendar month.
Seasonal index	Repository formula uses equal year weighting: 12 × mean of each year's monthly share. Values above 1 indicate an above-uniform monthly share.
Coefficient of Variation (CV)	Standard deviation divided by mean of monthly counts; higher values indicate more variation relative to the mean.
CR3	Combined share of all reported events or cases in the three highest months.
Gini coefficient	Mean-absolute-difference measure of inequality across the 12 monthly counts; higher values imply more concentrated monthly distribution.
Kendall's W / Spearman rho	Exploratory measures of agreement or rank-profile similarity across the four baseline years.
Kruskal–Wallis / permutation tests	Tests for differences in monthly distributions; statistical significance does not prove biological causation.

3.6 Significance and Interpretation Rules

Five disease-family tests were performed. A Bonferroni threshold of 0.01 is appropriate as a conservative reference when interpreting five simultaneous tests at an overall 0.05 level. Repository outputs report significant Kruskal–Wallis results for dengue, malaria and food poisoning, and non-significant results for ADD and chikungunya. Additional permutation and Kendall's W outputs differ by method; they are reported as exploratory evidence, not universal confirmation.

Kendall's W p-values use a Wilson–Hilferty chi-square approximation with only four annual profiles; the repository explicitly warns that exact permutation inference would be required for confirmatory claims. The report therefore uses evidence tiers and cautious wording rather than treating one p-value as a definitive biological result.

CHAPTER 3

Fieldwork Plan, Instruments and Ethics

3.7 Fieldwork Component in the Repository

The `field_work/` archive contains a full and short questionnaire, an interview guide, an observation checklist, a response template, a field-evidence register, a findings framework, a field-to-data linkage document, a presentation plan, and ethics/privacy guidance. The planned inquiry is intended to contextualize state surveillance patterns through the perspectives of residents and frontline actors; it is not intended to replace the secondary quantitative analysis.

Potential themes include awareness of seasonal timing, perceived environmental vulnerability, public-health interventions, notification pathways, surveillance literacy, data readiness, and community attitudes toward early-warning communication. The methodology recommends purposive/convenience sampling for contextual diversity, but its proposed target numbers are plans rather than realized sample sizes.

3.8 Ethics and Evidence Status

- Participation must be voluntary, based on informed consent, with the right to skip questions or withdraw.
- No patient charts, identifiable patient information, Aadhaar numbers, phone numbers, exact addresses or private clinical records should be collected.
- Photographs should document public infrastructure only; identifiable faces should not be captured without formal permission.
- Audio recording requires explicit permission; otherwise, use notes and report only anonymized summaries.
- No fabricated interviews, synthetic respondent counts, invented quotations or false field visits are allowed.

Evidence-status check: the repository version reviewed contains only the header row in `07_FIELD_EVIDENCE_REGISTER.csv`, and `06_FIELD_RESPONSE_TEMPLATE.csv` contains no completed respondent records. Student-supplied site-context images are now documented in Appendix D, but images alone do not provide questionnaire answers or establish interviews. Therefore, no participant counts, interview quotations, response percentages, or scored environmental-audit findings are claimed in this report.

CHAPTER 4

Observations, Mining and Analysis

4.1 Baseline Dataset Profile

The five-family baseline comprises 539 reported outbreak records over 2022–2025. Summing the family-level rows produces 20,955 reported cases and 266 reported deaths. Acute diarrheal disease has the largest reported case total, while dengue has the largest record count. The event count and the case count answer different questions: one very large event can contribute many cases without increasing event frequency by the same proportion.

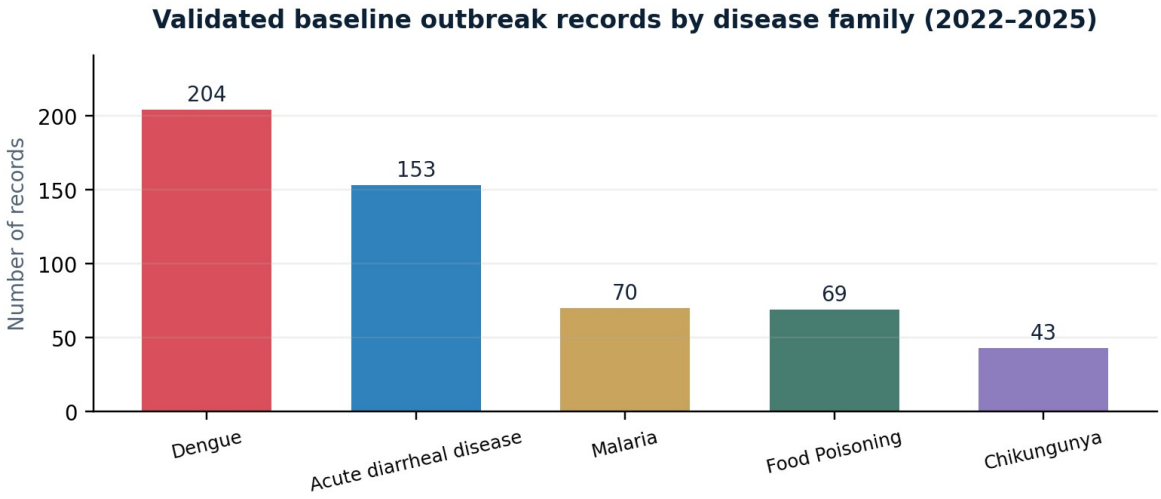


Figure 1. Number of validated reported outbreak records in the completed-year baseline. Source: Phase 5B monthly/yearly distribution files.

Disease family	Records	Reported cases	Reported deaths	Share of records
Dengue	204	3,144	93	37.8%
Acute diarrheal disease	153	9,050	93	28.4%
Malaria	70	2,133	51	13.0%
Food poisoning	69	5,927	29	12.8%
Chikungunya	43	701	0	8.0%
TOTAL	539	20,955	266	100.0%

Note: the table's total is the arithmetic sum of family rows. One Phase 5D report header states 21,955 baseline cases; the Phase 5B baseline and family-level totals support 20,955. The mismatch is explicitly flagged again in §4.5.

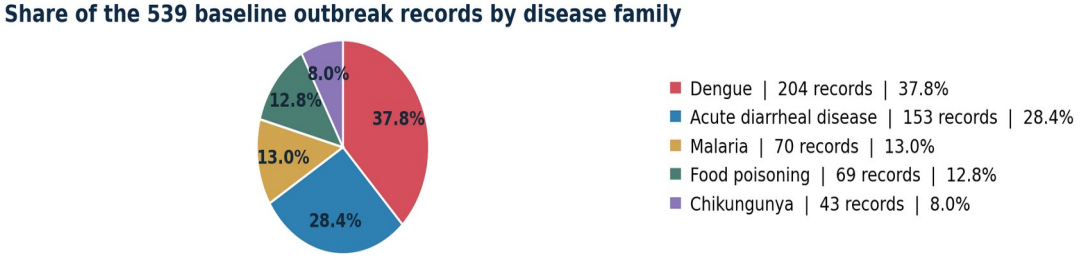


Figure 2. Share of the 539 baseline outbreak records by primary disease family. Percentages describe record frequency, not population incidence or case burden.

CHAPTER 4

Disease-Specific Seasonal Patterns

4.2 Monthly Distribution

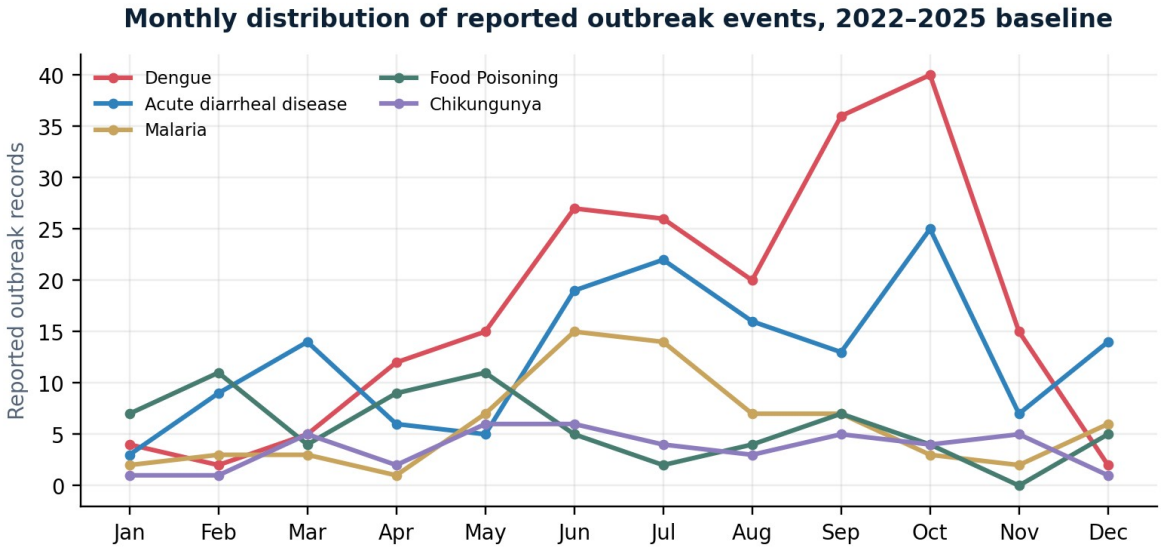


Figure 3. Monthly count profiles from the repository's 2022–2025 monthly distribution table; lines show event records, not individual cases.

Disease family	Observed timing in baseline	Interpretation
Dengue	June–October: 169/204 events (82.8%); October 40 records, September 36.	Strong late-monsoon/post-monsoon reporting concentration; peak month varies by year.
ADD	June–August: 66/153 events (43.1%); October also elevated (25 records).	Broader profile with a sustained mid-year rise and a secondary October elevation.
Malaria	May–July: 36/70 events (51.4%); June 15 and July 14 records.	Clearest contiguous early/monsoon-season reporting window.
Food poisoning	January–February 26.1%; April–May 29.0%; zero November records.	Bimodal and variable; not one continuous season.
Chikungunya	Top event months May and June (6 each, 14.0% each).	Diffuse, sparse event distribution without a stable contiguous window.

The patterns summarize government-notified outbreak records. They should not be read as direct measures of community incidence or as proof that a seasonal weather variable caused a disease peak.

CHAPTER 4

Concentration, Significance and Inter-Annual Stability

4.3 Concentration Metrics

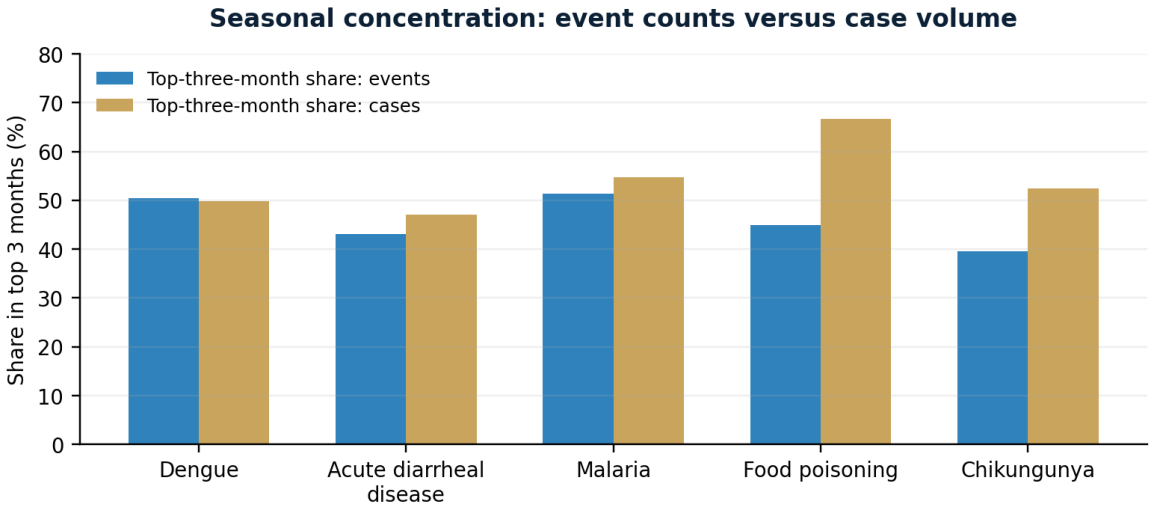


Figure 4. CR3 indicates the share of events or cases occurring in the three highest months; event concentration and case concentration can differ materially.

Family	Event CV	Event CR3	Case CV	Case CR3	Evidence tier
Dengue	0.736	50.5%	0.756	49.9%	A — strong descriptive
ADD	0.524	43.1%	0.609	47.1%	B — moderate descriptive
Malaria	0.753	51.4%	0.802	54.8%	B — moderate descriptive
Food poisoning	0.564	44.9%	1.160	66.7%	C — variable/context
Chikungunya	0.515	39.5%	0.783	52.5%	D — insufficient

Dengue and malaria have the highest event-level three-month concentration. Food poisoning stands out because its top-three-month share is much higher for cases (66.7%) than for event records (44.9%), showing the impact of several large reported outbreaks. Chikungunya has the lowest event CR3 and the least stable annual rank profile.

CHAPTER 4

Statistical Evidence and Cautious Interpretation

4.4 Statistical Results

Family	Kruskal–Wallis p-value	Kendall's W (p)	Interpretation in repository
Dengue	0.000219	0.405 (0.086)	Monthly distribution differs in KW test; annual profile only moderately concordant.
ADD	0.1321	0.530 (0.016)	KW not significant; highest annual rank concordance, interpreted as exploratory.
Malaria	0.005459	0.319 (0.229)	KW significant; core May–July window recurs but annual rank concordance is modest.
Food poisoning	0.005679	0.386 (0.108)	KW significant; pattern is event-driven and sensitive to high-volume records.
Chikungunya	0.3549	0.150 (0.832)	No significant monthly differences detected; sparsity limits inference.

Interpretation

Under a Bonferroni threshold of 0.01 across five family-level Kruskal–Wallis tests, dengue, malaria and food poisoning remain statistically significant; ADD and chikungunya do not. Separate repository permutation results are significant for more families, but the exact test procedures and their assumptions differ, so the report does not collapse them into one universal judgment.

Kendall's W results use an approximate chi-square p-value based on four yearly profiles. In particular, ADD's W = 0.530 and p = 0.016 are described as exploratory rank-concordance evidence, not conclusive proof of repeated biological seasonality. No statistical test establishes causation from the data described here.

CHAPTER 4

Outlier Sensitivity and District Concentration

4.5 Influential Records

Family	Influential record effect	Result after sensitivity check
Food poisoning	Three February records: Kolhapur 651, Parbhani 629, Solapur 335 cases (1,615 total).	February case share falls 36.5% → 12.7%; April becomes the largest case month (899; 20.8%).
ADD	A Nanded Week 06, 2024 event reports 1,000 cases.	February falls from the highest case month to a lower-ranked month; October/August rise relative to it.
Malaria	One Gadchiroli record reports 385 cases near late December / Week 01.	The December case peak is reduced; May–June become more prominent.
Dengue	No single record is marked as influential in the same outlier test.	October case peak remains after the exclusion check.
Chikungunya	Removing one Kolhapur 145-case record shifts case timing.	The apparent peak is sensitive to a single event because the event count is small.

4.6 Spatial Concentration

Malaria records are strongly concentrated in Eastern Vidarbha. Phase 5D records 43 of 70 baseline events (61.4%) in Gadchiroli (27) and Chandrapur (16), with both districts peaking in June and July. The report notes that this state-level timing profile should not be treated as geographically uniform. Pune accounts for 11 of 43 chikungunya records (25.6%), while dengue and ADD are reported across many more districts.

CHAPTER 4

Partial 2026 Comparison and Reconciliation Audit

4.7 Partial 2026 Through Week 32

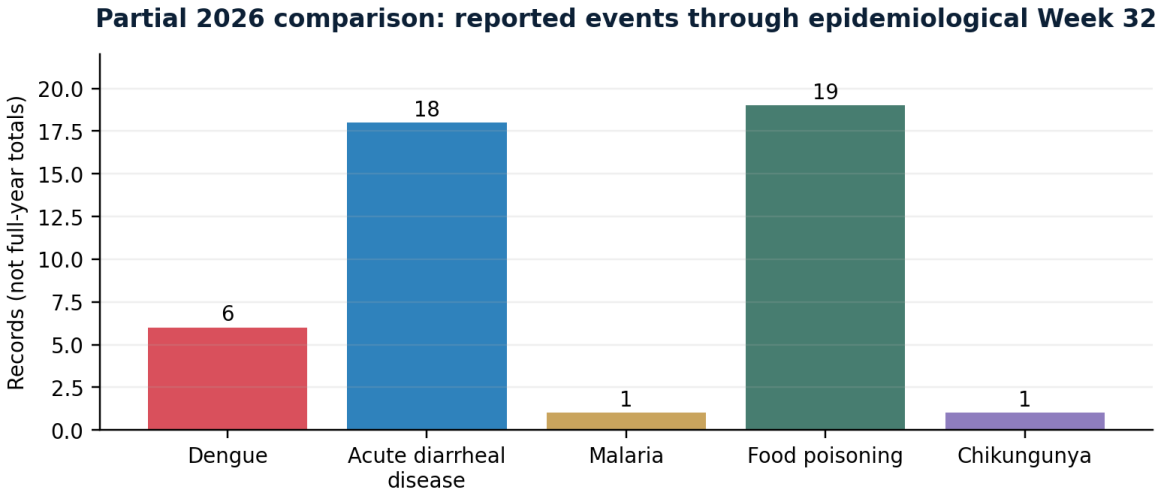


Figure 5. Selected family-level event counts through epidemiological Week 32 of 2026; this is not a full-year comparison.

Family	2026 events	Cases from family rows	Repository interpretation
Dengue	6	106	Early-year deficit cannot be interpreted as a mild year; late-season window is missing.
ADD	18	864	Close to the historical event-count comparison in the analysis report; label spelling shifted.
Malaria	1	65	One event reported in Gadchiroli; local reporting context should be examined.
Food poisoning	19	1,082	Higher event count than historical partial-window mean; case totals still event-sensitive.
Chikungunya	1	9	Only one record; too sparse for seasonal inference.
SUM OF FAMILY ROWS	45	2,126	Use for table-level arithmetic; see unresolved headline conflict.

4.8 Explicit Numerical Reconciliation

Two mismatches were detected in repository summary headers: (1) a Phase 5D header says 21,955 baseline cases, whereas Phase 5B and the sum of the five Phase 5D family totals equal 20,955; (2) a Phase 5D header reports 1,118 cases for 2026 Weeks 1–32, whereas the family-level 2026 table rows add to 2,126. This report uses the family-level totals shown above and

preserves both discrepancies as an unresolved documentation issue requiring source-code/output reconciliation.

CHAPTER 4

Repository, Website and Field Evidence Status

4.9 Repository and Website as Research Deliverables

The GitHub repository is organized into a source archive, data pipeline, fieldwork archive, and website. The pipeline preserves raw extraction, clean analytical data, validation records, disease-audit outputs, Phase 5B monthly/weekly profiles, Phase 5C statistics and outlier checks, and Phase 5D conclusions/limitations. This modular organization improves traceability because summary statements can be followed back to the underlying table or analysis phase.

The website documentation describes a Next.js/React/TypeScript application with D3-based visualizations and Tailwind styling. Its intended functions include disease-specific profiles, a seasonal analysis explorer, method transparency, and fieldwork/ethics documentation. It is a presentation layer for the analysis, not an independent data source; numerical statements should be governed by the validated dataset and frozen analytical outputs.

4.10 What Is and Is Not Empirically Complete

Component	Status inferred from repository reviewed	How this report treats it
Government-surveillance extraction and analysis	Documented outputs exist: CSVs, reports, validation and interpretation phases.	Main empirical source for findings.
Questionnaires / interview guide / checklist	Templates and protocols exist.	Appendix instruments; not counted as completed field evidence.
Field evidence register	Header only; no completed data rows.	No respondent totals, interview quotations or observation results claimed.
Website	Source and README document intended and implemented sections.	Described as dissemination layer, not a separate empirical dataset.

Visual materials supplied for this project are catalogued in Appendix D with conservative descriptions of visible setting and explicit provenance limitations. The primary qualitative strand remains incomplete in the available record because completed consented forms, interview notes/transcripts, and scored observation sheets were not supplied. This report therefore presents a finished analysis of the government-surveillance dataset and a documented visual field-context appendix, without representing unrecorded community responses as findings.

CHAPTER 5

Conclusion and Recommendations

5.1 Conclusions

The project demonstrates that the five selected disease families do not share a single common seasonal profile in the 2022–2025 government-notified outbreak record. Dengue shows strong late-year concentration, with 82.8% of events between June and October and the pooled monthly maximum in October. Malaria has the clearest contiguous window, with 51.4% of events in May–July, but its state profile is heavily influenced by Eastern Vidarbha. ADD shows a sustained June–August elevation with a secondary October rise. Food poisoning is temporally bimodal and its case volume depends heavily on a few large events. Chikungunya remains too sparse and variable for a stable seasonal claim.

The strongest lesson is methodological: an apparent seasonal peak can mean recurring outbreak events, high case volume from a few clusters, or a reporting artefact. Event counts, case counts, statistical significance, inter-annual repeatability and district distribution must be read together.

5.2 Recommendations

- Use the dengue late-season concentration to inform early planning and monitoring, while retaining district-level variation and annual shifts.
- Consider enhanced malaria surveillance in Eastern Vidarbha before and during the May–July window; do not generalize this regional profile to all districts.
- For ADD, monitor both the early/mid-monsoon window and October, while reviewing unusual reported death/case combinations before interpretation.
- For food poisoning, accompany headline case totals with event counts and sensitivity checks so mass-exposure events do not masquerade as seasonal recurrence.
- For chikungunya, increase data coverage and classification consistency before making strong timing claims.
- For ongoing reporting, publish reproducible aggregate tables, label partial years clearly, and automatically reconcile total rows with family totals.

These are planning suggestions based on descriptive surveillance evidence, not clinical advice or causal findings.

CHAPTER 5

Limitations and Future Work

5.3 Limitations

- Outbreak-only coverage: routine sporadic cases are not represented; the dataset cannot estimate population incidence or individual risk.
- Reporting differences: surveillance participation, investigation thresholds and reporting completeness can vary by district, year and disease.
- Incomplete current year: 2026 is limited to Weeks 1–32; major late-year windows are not observed.
- Missing weekly source files: the repository audit identifies missing 2023 Weeks 15, 51 and 52; these should not be replaced with fabricated zeroes.
- Label variation: suspected labels and spelling variants require normalization and can reflect differences in classification.
- Outlier distortion: several large reported event clusters materially change case-volume seasonality for food poisoning, ADD and malaria.
- Mortality anomalies: the December 2025 ADD cluster includes records with questionable case/death pairings; aggregate death totals should be treated with care.
- Limited years for inferential tests: four annual profiles provide only modest support for stable statistical inference; some test p-values are approximations.
- No primary field evidence in the evidence register: the fieldwork protocol is documented, but completed participant responses and observations are not verified in the repository reviewed.

5.4 Future Work

Next steps are to reconcile the two summary-total discrepancies against the current CSV/code; complete and verify the source archive for missing weeks; preserve the distinction between event and case metrics; extend the 2026 comparison after the year is complete; and, if approved, carry out ethical fieldwork with consented and anonymized evidence entries. A later version could add exact permutation-based inference, improved district-level normalization, meteorological covariates with verified temporal alignment, and formal reproducibility instructions. Such additions should be treated as future work until they are actually performed.

REFERENCES

References and Repository Sources

[1] National Centre for Disease Control (NCDC), Ministry of Health and Family Welfare, Government of India. Integrated Disease Surveillance Programme (IDSP) weekly outbreak surveillance reports for Maharashtra, 2022–2025; and 2026 Weeks 1–32. PDF source archive within the Field-Project repository.

[2] RizzyCoder19. Field-Project: Analysis of Seasonal Disease Patterns Using Government Health Data. GitHub repository, including README, source archive, data pipeline, validation reports, website and fieldwork archive. https://github.com/RizzyCoder19/Field-Project

[3] RizzyCoder19. `data_pipeline/analysis/NCDC_Maharashtra_CLEAN_ANALYTICAL.csv`. Clean analytical table and associated repository validation documentation, reviewed October 2026.

[4] RizzyCoder19. Phase 5B - Seasonal Analysis Report (2022–2025 Baseline), `data_pipeline/phase5B_seasonal_analysis/06_PHASE5B_ANALYSIS_REPORT.md`, with monthly distribution, seasonal index, yearly distribution and 2026 YTD comparison CSVs.

[5] RizzyCoder19. Phase 5C - Advanced Pattern & Statistical Analysis Report, `data_pipeline/phase5C_advanced_analysis/10_PHASE5C_ANALYSIS_REPORT.md`, including concentration, seasonal stability, peak timing, outlier sensitivity, district analysis and statistical testing outputs.

[6] RizzyCoder19. Phase 5D - Interpretation & Validation Report, `data_pipeline/phase5D_interpretation_validation/12_PHASE5D_ANALYSIS_REPORT.md`; and `09_FINAL_ANALYTICAL_CONCLUSIONS.md`, including evidence tiers, caveats, and unsupported-claims framework.

[7] RizzyCoder19. Fieldwork methodology, evidence register, qualitative findings framework, and ethics/privacy protocol, `field_work/05_FIELD_WORK_METHODOLOGY.md`, `field_work/07_FIELD_EVIDENCE_REGISTER.csv`, `field_work/09_FIELD_FINDINGS_FRAMEWORK.md`, and `field_work/13_FIELD_ETHICS_AND_PRIVACY.md`.

[8] “The Demographic and Seasonal Variation of Dengue Virus Infection in Gondia, A Tribal District in Maharashtra, India.” Identified in the repository source audit as an IJCPR paper (2026). Used as local context only; the present analysis does not merge that clinical series with IDSP outbreak notifications.

[9] R.P. Institute of Hospitality and Management, Field Project Website. Source/documentation reference in `website/README.md`; public site URL listed by the repository: https://fieldprojects.vercel.app

Source note: file names and analysis descriptions above follow repository documentation. Where complete bibliographic details were not exposed by the source-audit file, no author names or publication details have been invented.

APPENDIX A

Rapid Field Questionnaire and Evidence-Based Answer Key

The 20-item rapid instrument below follows the prepared field questionnaire in the repository. The supplied field response template contains no completed respondent rows, so the "evidence-based answer" is drawn only from the NCDC/IDSP analysis and the visible limits of the supplied photographs. It is not represented as a response from a resident, health worker, pharmacist or official.

1. Survey record metadata

Instrument / response options: Form code, date/time and locality/district.

Evidence-based answer: No completed respondent form or linked metadata was supplied; a respondent count or administration date cannot be stated from this file set.

2. Settlement type and district

Instrument / response options: Urban core, informal settlement/chawl, suburban, rural/peri-urban; district/locality.

Evidence-based answer: The supplied photographs show a public-health facility setting and a roadside/green verge screen capture associated by the user with the Vasai–Virar/Nalasopara project area. They do not establish a representative sample of settlements or districts.

3. Respondent category

Instrument / response options: Resident, pharmacist, ASHA/ANM, clinician/clinic staff, food vendor/school staff.

Evidence-based answer: Not established: no completed respondent forms are available.

4. Years residing or working locally

Instrument / response options: Under 1 year; 1–5; 6–10; over 10 years.

Evidence-based answer: Not established: no completed respondent forms are available.

5. Voluntary consent

Instrument / response options: Consent must be recorded as yes before the survey proceeds.

Evidence-based answer: The submitted materials contain the consent script and ethics protocol, but no completed consent records. This report therefore makes no claim about a respondent sample.

6. Months when illness increases

Instrument / response options: Pre-monsoon summer, early monsoon, late monsoon, post-monsoon, winter, or no pattern.

Evidence-based answer: The 2022–2025 surveillance baseline shows Dengue concentrated in June–October (169/204 records; 82.8%), Malaria in May–July (36/70; 51.4%), ADD in June–August (66/153; 43.1%), Food Poisoning in January–February and April–May, and Chikungunya without a stable contiguous window.

7. Disease families heard of or observed

Instrument / response options: Dengue, Malaria, ADD, Food Poisoning, Chikungunya.

Evidence-based answer: These five families form the primary analytical cohort in the project. This verifies their inclusion in the government-surveillance analysis, not whether any particular respondent observed them locally.

8. Highest peak months for mosquito-borne fevers

Instrument / response options: May–June, July–August, September–October, November–January, unsure.

Evidence-based answer: The state-level outbreak records place Dengue mainly in June–October; Malaria is most concentrated in May–July. Chikungunya is too sparse and unstable for a single reliable statewide peak window.

9. Seasonal change associated with diarrhoeal illness

Instrument / response options: Summer heat, first rains/pipeline concerns, waterlogging, winter, unsure.

Evidence-based answer: The records show a June–August rise in ADD outbreak-event frequency and a secondary October elevation. The dataset does not contain water-quality measurements that prove a pipeline or rainfall mechanism.

10. Localized food-poisoning events

Instrument / response options: Hot summer, community meal/gathering, school canteen/mess, or none.

Evidence-based answer: Three high-case February records in Kolhapur, Parbhani and Solapur total 1,615 reported cases. This supports the presence of large reported clusters in the surveillance archive, but does not identify a common meal or respondent-level observation.

11. Most visible environmental risk

Instrument / response options: Stagnant drains, potholes, construction water, uncovered storage, none.

Evidence-based answer: The facility photos document access and service environments. The available imagery does not systematically assess household storage, drain condition or a 50–100 m environmental transect, so no risk factor can be ranked from the photos alone.

12. Main drinking-water source

Instrument / response options: Municipal tap, tank, standpost/tanker, packaged/RO, well/borewell.

Evidence-based answer: Not established from the supplied photos or government outbreak dataset.

13. Visibility/timing of prevention activity

Instrument / response options: Pre-monsoon, reactive after cases, rare/complaint-led, never observed.

Evidence-based answer: Not established. No completed respondent answer, dated observation checklist or intervention log was supplied.

14. Visibility of government health warnings

Instrument / response options: High, medium, low/none.

Evidence-based answer: The supplied images do not provide a systematic poster/announcement inventory. No rating can be assigned.

15. First point of care for fever or vomiting

Instrument / response options: Private GP, chemist, public facility, home remedies.

Evidence-based answer: Not established: no completed responses on healthcare-seeking behaviour were provided.

16. Awareness of IDSP/NCDC

Instrument / response options: Aware or not familiar.

Evidence-based answer: Not established: the secondary dataset is published by the surveillance system, but does not measure community awareness.

17. Do official outbreak reports capture every illness?

Instrument / response options: Most cases, severe clusters only, miss mild/private cases, unsure.

Evidence-based answer: Methodologically, the NCDC/IDSP outbreak file records notified and investigated outbreak clusters; it is not a census of all individual illnesses. It cannot estimate population incidence or the number of unreported mild cases.

18. Usefulness of a weekly health alert

Instrument / response options: Very useful, somewhat useful, not useful.

Evidence-based answer: Not measured because no respondent answers were supplied. A weekly alert is a reasonable design recommendation to test in future fieldwork, not a demonstrated community preference.

19. One practical prevention step

Instrument / response options: Open response.

Evidence-based answer: Based on the surveillance timing, preparedness should be timed before the observed windows: early preparation for Malaria (May–July), sustained Dengue surveillance (June–October), and water/food-safety attention around the observed ADD and Food Poisoning clusters. This is an analysis-derived recommendation, not a respondent quote.

20. Investigator checklist

Instrument / response options: Stagnant water visible; anti-larval/fogging marks visible; waste accumulation visible.

Evidence-based answer: The response template and evidence register contain no completed scored observation sheets. The photos are therefore presented as visual context and are not converted into environmental risk scores.

How to read this appendix

The answer key answers the research questions only where the government-surveillance records or supplied images support a response. “Not established” is a result of missing primary records, not an unanswered placeholder. No survey totals, respondent percentages, quotations, consent outcomes or environmental scores have been fabricated.

APPENDIX B

Interview Guide and Observation Checklist

B1. Semi-Structured Interview Prompts

- What seasonal changes in reported illness do you observe in your role, and what months are most noticeable?
- How are suspected local outbreaks identified, reported and escalated to public-health authorities?
- What barriers can delay notification (access, workload, definitions, connectivity, awareness or other factors)?
- What prevention and risk-communication activities are commonly undertaken before or during monsoon periods?
- How are outbreak information and prevention guidance communicated to local residents?

- What data would help local teams prepare earlier, and what are the risks of misinterpreting an early seasonal signal?

B2. Environmental Observation Checklist

Item to observe	Record only visible, non-identifying information
Standing water / containers	Location code, approximate context, whether water is visible; no private-home identifiers.
Drainage and waterlogging	Public infrastructure condition, flow or blockage observations, photo only if permitted.
Public-health information	Presence, topic, language and visible condition of public notices/signage.
Vector-control activity	Visible public activity or notice, date/time if known; do not infer effectiveness from presence alone.
Food/water safety context	Public facility context or visible sanitation condition; no accusations or causal claims.
Evidence handling	Unique evidence ID, date, general locality, consent status if applicable, source and verification status.

Boundary: an observation can describe a visible setting but cannot establish that a specific drain, household, shop, or intervention caused a specific disease case. Triangulation would require properly collected, ethical, verifiable evidence.

B3. Core Semi-Structured Interview Questions

1. How does the burden of acute illness change across the year, and what signs suggest the start of a local increase?

2. Does the late-monsoon Dengue pattern and the earlier May–July Malaria pattern match what you encounter locally? Please distinguish personal observation from information heard elsewhere.

3. When do vomiting or diarrhoeal illnesses become most noticeable, and what local water or food conditions, if any, are observed at the same time?

4. What usually triggers recognition that several nearby illnesses may form a local cluster?

5. How can information about a suspected cluster travel from a household or clinic to municipal or IDSP surveillance?

6. Which points can delay reporting, such as access, definitions, workload, test availability or connectivity?

7. How are suspected and confirmed disease labels distinguished in everyday reporting?

8. What mosquito-control, water-safety or food-safety actions are generally used before and during monsoon periods?

9. How are residents informed about prevention actions, warning signs and when to seek qualified care?

10. What visible public-health information or prevention materials are generally available in this setting?

11. Would a simple weekly local alert or seasonal chart be useful to staff or residents, and what risks might it create if misunderstood?

12. What single change would most improve early detection, public communication or data quality in this area?

Interview evidence status: the repository contains the interview guide but no completed transcripts or interview notes. These are the approved interview prompts, not interview findings.

APPENDIX C

Metrics, Data-Quality Register and Project Planner

C1. Metric Reminder

Metric	Working definition	Use / caution
Seasonal Index	12 × mean within-year monthly share (equal weighting across completed years).	Above 1 indicates above-uniform month share; keep index method consistent.
CV	Standard deviation of monthly counts ÷ mean monthly count.	Higher = more relative variation; can be affected by zero months.
CR3	Sum of counts in three highest months ÷ total annual-period count.	Compute separately for outbreak events and reported cases.
Gini	Mean-absolute-difference inequality across 12 monthly counts.	Higher = more monthly concentration; formula must be stated.
Kendall’s W	Concordance among the four annual month-rank profiles.	Treat approximate p-values as exploratory with only four years.

Metric	Working definition	Use / caution
Kruskal–Wallis	Nonparametric test of differences among monthly distributions.	Statistical difference does not identify cause or prove a forecasting relation.

C2. Reconciliation Register

Issue	Conflict in repository	Treatment in this report	Verification treatment in this report
Baseline case total	Phase 5D header: 21,955; Phase 5B and family rows: 20,955.	Use 20,955, the sum of five family rows; flag mismatch.	Reconcile header to current CSV / aggregation script.
2026 partial cases	Phase 5D header: 1,118; family rows: 106 + 864 + 65 + 1,082 + 9 = 2,126.	Use family-level sum 2,126 in the table; flag mismatch.	Re-run YTD aggregation and update summary header.
Dengue index values	Different prose summaries disagree on exact September/October index values.	Use stable monthly record totals (Oct 40, Sep 36) in the main result.	Recompute from the canonical seasonal-index CSV.
ADD deaths	December 2025 records contain repeated high death counts alongside very small case totals.	Retain as source-faithful rows, but do not make biological claims.	Check source PDFs and extraction field mapping.
Fieldwork evidence	Evidence register contains headers only.	Do not report respondent numbers or interview findings.	Complete approved fieldwork and add verified anonymized records if required.

C3. Research Project Planner and Repository Milestones

Stage	Task	Evidence to retain
1	Confirm source and analysis scope	Research questions and source list
2	Check and validate weekly PDFs	Validation log and gap notes
3	Normalize disease labels	Mapping table with decisions
4	Recompute seasonal outputs	CSV tables and reproducible code
5	Audit discrepancies and outliers	Reconciliation / sensitivity register
6	Optional ethical field inquiry	Consent evidence and anonymized register entries
7	Revise report and figures	Versioned report and page-level QA
8	Guide/HOD review and submit	Approved certificate signatures and official stamp

C4. Guide Interaction Diary and Academic Review Record

No dated guide-interaction diary or meeting minutes were supplied with the repository and attachments. To avoid attributing meetings or statements to Prof. Angelin without a record, the following is a topic-based review register tied to project artefacts, not a claim that each discussion occurred on a particular date.

Review area	Documented review focus
Research question and scope	Defined Maharashtra outbreak-notification analysis, 2022–2025 baseline and a separate 2026 Weeks 1–32 comparison.
Source and processing pipeline	Reviewed the weekly NCDC/IDSP archive, extraction outputs, disease-family mapping and data-quality documentation.
Statistical findings	Reviewed seasonal concentration, Kendall’s W, Kruskal–Wallis tests, district concentration and outlier sensitivity.
Fieldwork and ethics	Reviewed the supplied site photographs and the questionnaire, interview, observation and ethics templates; no respondent records were supplied.
Report quality and submission	Aligned the report with the supplied Mumbai University guideline for A4, Times New Roman, 12-point body text, 1.5 spacing, bottom-centred page numbering, references and appendices.

APPENDIX D

Field Site Photographic Documentation

The following student-supplied images document visual context associated with public-health facilities and access environments in the Vasai–Virar/Nalasopara area. Captions report only visible features; image dates and capture provenance are not independently verified for every item. A screenshot is identified as such. The crowd image is privacy-softened in this report copy.



Figure D1. Covered public-hospital corridor with waiting/seating and a queue-like gathering. The image supports a description of the service-use setting only; it does not provide counts of patients or a measure of facility performance.



Figure D2. Indoor service/reception corridor image supplied for context. Visible service-counter layout is described conservatively; the embedded timestamp reads 22 March 2022, so it is not treated as evidence of a visit on the report date.

Field-evidence boundary: these images are visual material only. No questionnaire answer, interview statement, sample size, or scored environmental observation has been inferred from them.

APPENDIX D | FIELD SITE PHOTOGRAPHIC DOCUMENTATION



Figure D3. Exterior view of a municipal public-health facility with a covered forecourt and entry approach. The visible structure provides access-setting context; no claim about services, staffing, or current condition is inferred.



Figure D4. Second exterior view of a municipal facility entrance and covered approach. The image is used to illustrate the facility frontage and entry area, not as a disease-risk measure.

Field-evidence boundary: these images are visual material only. No questionnaire answer, interview statement, sample size, or scored environmental observation has been inferred from them.

APPENDIX D | FIELD SITE PHOTOGRAPHIC DOCUMENTATION



Figure D5. Screen capture of a map/street-view interface showing a covered facility approach and adjacent roadside/green verge. This is identified as a screen capture, not a direct camera photograph; vegetation or a water feature alone does not prove a breeding site.



Figure D6. Clinical/procedure-room image supplied as contextual visual material. Equipment is visible, but the image does not establish current operational status, patient use, or a connection with any reported outbreak.

Field-evidence boundary: these images are visual material only. No questionnaire answer, interview statement, sample size, or scored environmental observation has been inferred from them.